

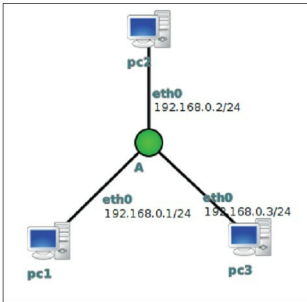


Figure 1: A network with three hosts interconnected via a network hub

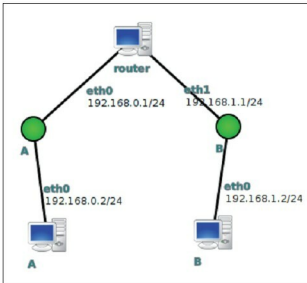


Figure 2: A network with a router to bridge two families of networks

We can shut down the machine from outside the virtual terminal using the command `vhalt [virtual host name]`. If something goes wrong and needs the virtual machine to be forcefully shut down, use the command `vcrash [hostname]`.

The `—con0` argument for the `vstart` command will help you in specifying the terminal to be used for a VM. To use the same terminal in which commands are typed, use `—con0=this` or if you want Konsole as the terminal, use `—con0=konsole`.

Some network manipulation/config commands

- The `ping` command is to check connectivity. We can ping another machine on a network using its IP address/hostname and check the connectivity by

looking at the command output. For example: `ping google.com`

- `ifconfig` is used to set the IP address for an interface. To set the IP address of `eth0`, just use `ifconfig eth0 <ip address> netmask <netmask>`
- The `route` command can be used to set the routes for an IP network. For example, `route add default gw 192.168.0.1 dev eth0` sets 192.168.0.1 as the default gateway for the interface as `eth0`.
- You can use the `tcpdump` command for packet sniffing. With this command, you can monitor the network data packets transfer through a network interface. For example, `tcpdump -i eth0`
- You can use `traceroute` to trace the packet route. This means that you can find out the gateways through which a packet travels to reach its destination. For example, `traceroute mec.ac.in`

Hands-on network emulation

We will now look at how to emulate simple networks using Netkit. A network with three hosts interconnected via a network hub is shown in Figure 1.

It is very easy to set up a network of machines connected through a common network hub. Execute the following commands:

```
slyrux@grubox-$ vstart pc1 -eth0=A
slyrux@grubox-$ vstart pc2 -eth0=A
slyrux@grubox-$ vstart pc3 -eth0=A
```

These three commands will create three terminals, namely Virtual Console #0 (pc1), Virtual Console #0 (pc2), Virtual Console #0 (pc3).

Let's list the details of active virtual machines using the command, `vlist`.

```
slyrux@grubox-$ vlist
USER      VHOST      PID      SIZE  INTERFACES
slyrux    pc1        4485     22852
slyrux    pc2        4944     22852
slyrux    pc3        5503     22852
```

```
Total virtual machines: 3 (you), 3 (all users).
Total consumed memory: 68556 KB (you), 68556 KB (all users).
slyrux@grubox-$
```

Now we can set IP addresses for each of the machines using `ifconfig`:

```
pc1# ifconfig eth0 192.168.0.1
pc2# ifconfig eth0 192.168.0.2
pc3# ifconfig eth0 192.168.0.3
```

Try to ping between these machines—for example, from pc1 (with IP address 192.168.0.1), ping 192.168.0.2. A router is a network device used to bridge two